

ECE 3441 Spring 2025

Quiz 1

VERSION B

- Closed book, closed notes
- No cheat sheet is allowed.
- Only four function calculators are allowed.
- Show all details of your work to obtain credit (The answer alone will not suffice).
- You may not talk during the quiz to others or to yourself.
- You have 60 minutes.

Please do not open booklet unless instructed to do so, or substantial penalty will result.

Rishi Dasari */* Revised */*
SOLUTIONS *2025/02/24*

First Name Last Name, Peoplesoft ID, Date (YYYY/MM/DD)

Question	Points	Maximum
1		30
2		20
3		20
4		15
5		25

Total: _____/(100 possible points + 10 bonus points)

Question #1:

- a) Convert the following decimal number to binary (10 points).

$$625.0625_{10}$$

$$\begin{array}{rcl}
 625 : 1 & 0.0625 & \\
 312 : 0 & 0 : 0.1250 & \\
 156 : 0 & 0 : 0.2500 & \\
 78 : 0 & 0 : 0.5000 & \\
 39 : 1 & 1 : 1.0000 & \\
 19 : 1 & & \\
 9 : 1 & & \\
 4 : 0 & & \\
 2 : 0 & & \\
 1 : 1 & &
 \end{array}
 \Rightarrow \boxed{10,0111,0001.0001_2}$$

- b) Convert the value in a) to hexadecimal (two digits after the radix point). (5 points)

$$\begin{array}{cccccc}
 0010 & , & 0111 & , & 0001 & . & 0001 & , & 0000 \\
 \hline
 2 & & 7 & & 1 & & 1 & & 0
 \end{array}
 \Rightarrow \boxed{271.10_{16}}$$

- c) Using **8-bit 2's-complement** binary arithmetic, perform the subtraction below and convert the answer to decimal. Remember: first convert each decimal number to its 8-bit 2's-complement binary representation before performing the operation. (15 points)

$$75_{10} - 24_{10}$$

$$\begin{array}{rcl}
 75 : 1 & & \\
 37 : 1 & & \\
 18 : 0 & & \\
 9 : 1 & & \\
 4 : 0 & & \\
 2 : 0 & & \\
 1 : 1 & & \\
 \hline
 0 & & \\
 24 : 0 & & \\
 12 : 0 & & \\
 6 : 0 & & \\
 3 : 1 & & \\
 1 : 1 & & \\
 0 & &
 \end{array}
 \begin{array}{l}
 75_{10} \Rightarrow 0100,1011_2 \\
 -24_{10} \Rightarrow 0001,1000 \\
 \quad \quad \quad \rightarrow 1110,0111 \\
 \quad \quad \quad \rightarrow 1110,1000_2 \\
 \quad \quad \quad \begin{array}{r}
 0100,1011 \\
 + 1110,1000 \\
 \hline
 1,0011,0011 \\
 \Rightarrow \boxed{0011,0011_2} \\
 = 1 + 2 + 16 + 32 = \boxed{51_{10}}
 \end{array}
 \end{array}$$

Question #2:

For each expression, identify what theorem/law can be used to simplify it **in one step**. Then, apply the theorem to the expression. Remember: you can treat expressions in parentheses like whole variables in the theorems. (20 points)

a) $AB'C + BC$

$$= AC + BC \quad : \text{ Elimination (2-3, 3)}$$

b) $ABC + B'C$

$$= AC + B'C \quad : \text{ Elimination (2-3, 3)}$$

c) $VWX'Z' + W'VYX' + ZYX'V$

~~$$= VX'WZ' + VX'W'Y + VX'YZ + VX'YZ' + VX'YW \quad : \text{ Consensus (2-3, 6)}$$~~

$$= VX'(WZ' + W'Y + YZ) \quad : \text{ Distributive}$$

$$= VX'(WZ' + W'Y + YZ + YZ') \quad : \text{ Consensus}$$

$$= VX'(WZ' + W'Y + Y) \quad : \text{ Uniting}$$

$$= VX'(WZ' + Y) \quad : \text{ Absorption}$$

d) $(CAB' + E)(CAB' + DF')$

$$= CAB' + DEF' \quad : \text{ Distributive (2-4, 8D)}$$

Question #3:

- a) Eliminate the XOR and then factor to obtain the minimum POS of the following. (10 points)

Let $F = Y'Z + (WX \oplus Y)$

$$F = Y'Z + WXY' + W'Y + X'Y$$

~~$$F = F'' = ((Y+Z')(W'+X'+Y)(W+Y')(X+Y'))'$$~~

~~$$= (((W'+X')Z' + Y)(WX + Y'))'$$~~

~~$$= ((W'Z' + X'Z' + Y)(WX + Y'))'$$~~

~~$$= (W'Y'Z' + X'Y'Z' + WXY)'$$~~

$$= (W + Y + Z)(X + Y + Z)(W' + X' + Y')$$

/* Expected Process on next page */

- b) Prove or disprove the following: $W \oplus XY = (W \oplus X)(W \oplus Y)$. (10 points)

$$W \oplus XY = W'(XY) + W(XY)'$$

$$= W'XY + WX' + WY'$$

$$(W \oplus X)(W \oplus Y) = (WX' + W'X)(WY' + W'Y)$$

$$= WX'Y' + W'XY$$

$$W'XY + WX' + WY' \neq WX'Y' + W'XY$$

$$\therefore W \oplus XY \neq (W \oplus X)(W \oplus Y)$$

/* Counterexample(s) would also be fine */

	WX			
	00	01	11	10
00	0	0	1	0
01	1	1	1	1
11	1	1	0	1
10	1	1	0	1
YZ				

POS: $(W' + X' + Y')$
 $\dots (W + Y + Z)$
 $\dots (X + Y + Z)$

W \ XY	00	01	11	10
0	0	0	1	0
1	1	1	0	1

W \ XY	00	01	11	10
0	0	0	1	0
1	1	0	0	0

Question #4: 4 in this version

Simplify the following to only 3 terms using only the consensus theorem or its dual. (15 points)

a) $XY'Z + \underline{W'YZ} + \underline{W'XZ} + X'YZ + \underline{W'XY'}$

$$= \boxed{XY'Z + W'YZ + X'YZ + W'XY'}$$

/★ continued from previous page ★/

3.) a.) $Y'Z + (WX \oplus Y)$

$$= Y'Z + WXY' + (WX)'Y : \text{XOR}$$

$$= Y'Z + WXY' + (W' + X')Y : \text{DeMorgan's}$$

$$= (WX + Z)Y' + (W' + X')Y : \text{Distributive}$$

$$= (WX + Z + Y)(W' + X' + Y') : \text{MOAF}$$

$$= (W + Y + Z)(X + Y + Z)(W' + X' + Y') : \text{Distributive}$$

} 4 pts.

} 6 pts.

Question #5:

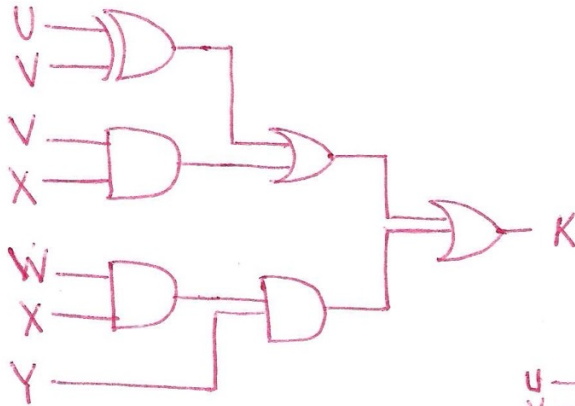
- a) Simplify and sketch a circuit that uses no more than six 2-input logic gates (10 points).

$$K = WXY + UV' + \underline{UX} + \underline{VX} + \underline{U'V}$$

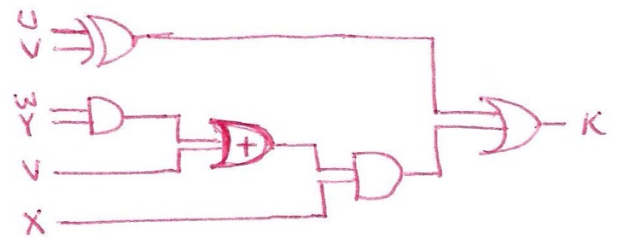
$$= WXY + \cancel{VX} + UV' + U'V \quad : \text{Consensus}$$

$$= WXY + VX + (U \oplus V) \quad : \text{XOR}$$

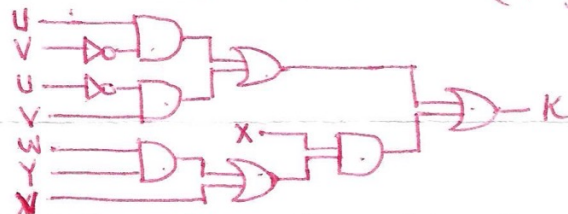
$$= (WY + V)X + (U \oplus V) \quad : \text{Distributive}$$



Alternative: (5 gates)



Alternative: (9 gates)



~~$(WY + V)X$~~ /* XOR not required */
/* Up to 9 (or 10) is fine */

- b) For the circuit implemented in Part a), calculate the maximum pull-up resistance that can be used for input B, using the provided data. (15 points)



Table 1: Input and Output Parameters for Various TTL Chips

		74LS00 (NAND)	74LS02 (NOR)	74LS04 (NOT)	74LS08 (AND)	74LS32 (OR)	74LS86 (XOR)
V_{CC}	NOM	5 [V]					
V_{IH}	MIN	2 [V]	2 [V]	2 [V]	2 [V]	2 [V]	2 [V]
V_{IL}	MAX	0.8 [V]	0.8 [V]	0.8 [V]	0.8 [V]	0.8 [V]	0.8 [V]
V_{OH}	MIN	2.5 [V]	2.4 [V]	2.4 [V]	2.7 [V]	2.7 [V]	2.7 [V]
	TYP	3.4 [V]	3.4 [V]	3.4 [V]	3.4 [V]	3.4 [V]	3.4 [V]
V_{OL}	TYP	0.35 [V]	0.2 [V]	0.2 [V]	0.35 [V]	0.35 [V]	0.35 [V]
	MAX	0.5 [V]	0.4 [V]	0.4 [V]	0.5 [V]	0.5 [V]	0.5 [V]
I_{OH}	MAX	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]
I_{OL}	MAX	8 [mA]	16 [mA]	16 [mA]	8 [mA]	8 [mA]	8 [mA]
I_{IH}	MAX	20 [μA]	40 [μA]	20 [μA]	20 [μA]	20 [μA]	40 [μA]
I_{IL}	MAX	-0.4 [mA]	-1.6 [mA]	-0.4 [mA]	-0.4 [mA]	-0.4 [mA]	-0.8 [mA]

SCRATCH

V is fed to an XOR and AND

$$\text{XOR: } R_{PU} < \left(\frac{(5[V]) - (2[V])}{(40 [\mu A])} = 75 [k\Omega] \right)$$

$$\text{AND: } R_{PU} < \left(\frac{(5[V]) - (2[V])}{(20 [\mu A])} = 150 [k\Omega] \right)$$

/★ Regardless of circuit in part a.) must take statement by student when considering work and answer.

If multiple are calculated, lowest must be chosen. ★/

Uniting theorems:

1. $XY + XY' = X$

Absorption theorems:

2. $X + XY = X$

Elimination theorems:

3. $X + X'Y = X + Y$

Duality:

4. $(X + Y + Z + \dots)^D = XYZ \dots$

Theorems for multiplying out and factoring:

5. $(X + Y)(X' + Z) = XZ + X'Y$

Consensus theorems:

6. $XY + YZ + X'Z = XY + X'Z$

1D. $(X + Y)(X + Y') = X$

2D. $X(X + Y) = X$

3D. $X(X' + Y) = XY$

4D. $(XYZ \dots)^D = X + Y + Z + \dots$

5D. $XY + X'Z = (X + Z)(X' + Y)$

6D. $(X + Y)(Y + Z)(X' + Z) = (X + Y)(X' + Z)$

Operations with 0 and 1:

1. $X + 0 = X$

2. $X + 1 = 1$

Idempotent laws:

3. $X + X = X$

Involution law:

4. $(X')' = X$

Laws of complementarity:

5. $X + X' = 1$

Commutative laws:

6. $X + Y = Y + X$

Associative laws:

7. $(X + Y) + Z = X + (Y + Z)$
 $= X + Y + Z$

Distributive laws:

8. $X(Y + Z) = XY + XZ$

DeMorgan's laws:

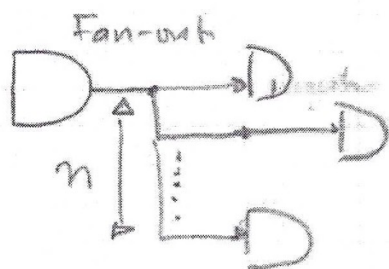
9. $(X + Y)' = X'Y'$

Exclusive OR

$X \oplus Y = X'Y + XY'$

Equivalence

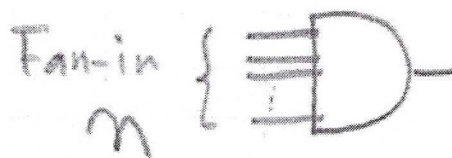
$(X \equiv Y) = XY + X'Y'$



Fan-out $L = n$

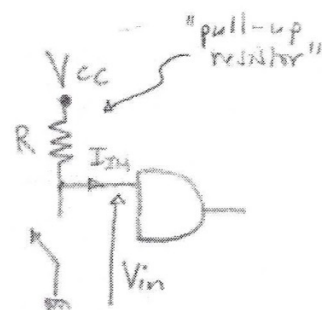
Fan-in $H = m$

Fan-out $= \min(m, n)$



$m < \frac{I_{OHmax}}{I_{IHmax}}$

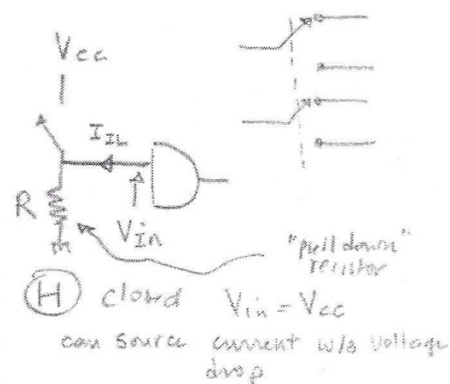
$m < \frac{I_{OLmax}}{I_{ILmax}}$



(L) closed $V_{in} = 0$ (L)
 can sink current without generating voltage

(H) open
 $V_{in} = V_{cc} - I_{IH}R$
 $V_{in} > V_{IHmin}$

$V_{cc} - I_{IH}R > V_{IHmin}$
 $\boxed{\frac{V_{cc} - V_{IHmin}}{I_{IHmax}} > R}$



(H) closed $V_{in} = V_{cc}$
 can source current w/o voltage drop

(L) open
 $V_{in} = RI_{IL}$
 $V_{in} < V_{ILmax}$

$RI_{IL} < V_{ILmax}$

$\boxed{R < \frac{V_{ILmax}}{I_{ILmax}}}$

